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Theoretical Study on the Improvement of Luminescence Efficiency in β–Estradiol Matrix

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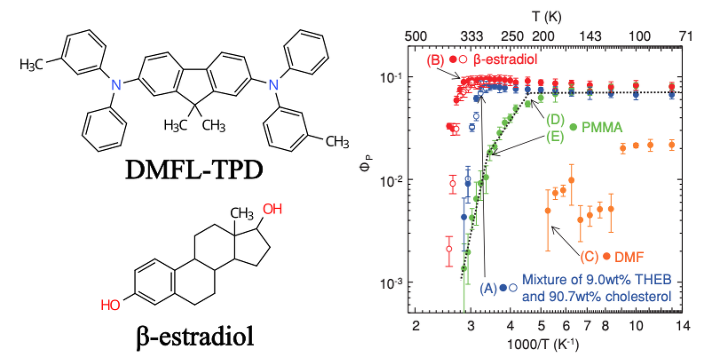
Background: Room Temperature Phosphorescence

Room temperature phosphorescence is useful for OLED, bioimaging etc...

Hirata *et al.* reported that DMFL–TPD exhibits enhanced phosphorescence efficiency when doped in a β–estradiol matrix [*1].



The theoretical reason for such improvement has been unclear.

[*1] S. Hirata *et al.*, *Adv. Funct. Mater.* **2013**, *23*, 3386.

Background: Calculation of Dopant in Matrix

In large systems, computational cost becomes prohibitive.

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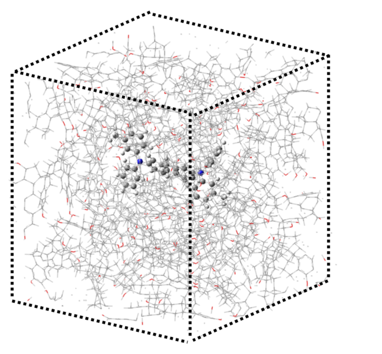
It is difficult to obtain structures through quantum chemical calculations.



In recent years, the advent of Matlantis[*2] (DFT–based machine learning potentials) has enabled high–speed quantum chemical calculations.

Goal

To elucidate the mechanism of nonradiative decay from T_1 contributing to the improvement of phosphorescence efficiency of DMFL–TPD in β–estradiol matrix.

[*2] https://matlantis.com/

Methods

1. Obtain the doped matrix structure under 1 atm at 300 K from NPT–MD simulations using Matlantis (PFP version 7.0.0[*3]).
2. DMFL–TPD and its adjacent β–estradiol are included in the QM region, and ground– and excited–state calculations are conducted using ONIOM method implemented in Gaussian Rev. C02.
3. Calculations of DMFL–TPD in DMF solution are also carried out for comparison.
4. For both systems, spin–orbit coupling calculations are performed to obtain the mixed spin states and analyses of vibronic couplings are conducted by using in–house codes.

Calculation levels

- in DMF solution: ωB97X–D/6-31G(d,p), PCM
- in the β–estradiol matrix: ONIOM(ωB97X–D/6-31G(d,p):UFF)

[*3] S. Takamoto *et al.*, *Nature Commun.* **2023**, *13*, 2991.

Theory: Rate Constant of Nonradiative decay[*4]

Nonradiative decay can be considered as the transition between mixed–spin states[*5]. Mixed–spin states whose main component are T_1 and S_0 are:

$$\left|\widetilde{\mathrm{T}}_1\right\rangle = \sum_{n=1} C^{\widetilde{\mathrm{T}}_1}_{\mathrm{S}_n}\left|\mathrm{S}_n\right\rangle + \sum_{m=1} C^{\widetilde{\mathrm{T}}_1}_{\mathrm{T}_m}\left|\mathrm{T}_m\right\rangle, \quad \left|\widetilde{\mathrm{S}}_0\right\rangle = \sum_{n=1} C^{\widetilde{\mathrm{S}}_0}_{\mathrm{S}_n}\left|\mathrm{S}_n\right\rangle + \sum_{m=1} C^{\widetilde{\mathrm{S}}_0}_{\mathrm{T}_m}\left|\mathrm{T}_m\right\rangle. \quad (1)$$

The rate constant of nonradiative decay is

$$k^{\widetilde{\mathrm{T}}_1\to\widetilde{\mathrm{S}}_0}_{\mathrm{nr}} = \sum_{\alpha,\beta} k^{\widetilde{\mathrm{T}}_1\to\widetilde{\mathrm{S}}_0}_{\mathrm{nr},\alpha\beta} = \sum_{\alpha,\beta}\frac{2\pi}{\hbar}V_\alpha^{\widetilde{\mathrm{T}}_1-\widetilde{\mathrm{S}}_0}V_\beta^{\widetilde{\mathrm{S}}_0-\widetilde{\mathrm{T}}_1}\Theta_{\alpha\beta} \quad (2)$$

where $V_\alpha^{\widetilde{\mathrm{T}}_1-\widetilde{\mathrm{S}}_0} := \left\langle \widetilde{\mathrm{T}}_1 \middle| \left(\frac{\partial \hat{H}_{\mathrm{e}}}{\partial Q_\alpha}\right)_{\mathbf{0}} \middle| \widetilde{\mathrm{S}}_0 \right\rangle$ ($\hat{H}_{\mathrm{e}}$: electronic Hamiltonian, Q_α: normal coordinate along vibrational mode α) is vibronic coupling constant (VCC) along vibrational mode α:

$$V_\alpha^{\widetilde{\mathrm{T}}_1-\widetilde{\mathrm{S}}_0} = \sum_{n,n'}\left(C^{\widetilde{\mathrm{T}}_1}_{\mathrm{S}_n}\right)^* C^{\widetilde{\mathrm{S}}_0}_{\mathrm{S}_{n'}} V_\alpha^{\mathrm{S}_n-\mathrm{S}_{n'}} + \sum_{m,m'}\left(C^{\widetilde{\mathrm{T}}_1}_{\mathrm{T}_m}\right)^* C^{\widetilde{\mathrm{S}}_0}_{\mathrm{T}_{m'}} V_\alpha^{\mathrm{T}_m-\mathrm{T}_{m'}}, \quad (3)$$

[*4] W. Ota, M. Uejima, N. Haruta, T. Sato, *Bull. Chem. Soc. J.* **2024**, *97*, uoad020. [*5] T. Azumi, *Chem. Phys. Lett.* **1972**, *17*, 211.

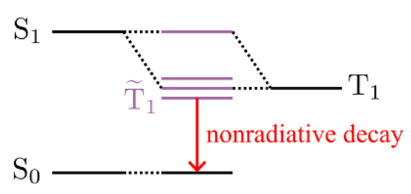
and $\Theta_{\alpha\beta}$ is the nuclear part which depends on Boltzmann factor $P_{\widetilde{\mathrm{T}}_1\nu}(T)$

$$\Theta_{\alpha\beta} = \sum_{\nu,\nu'} P_{\widetilde{\mathrm{T}}_1\nu}(T)\left\langle \chi_{\widetilde{\mathrm{S}}_0\nu}\middle|Q_\alpha\middle|\chi_{\widetilde{\mathrm{T}}_1\nu}\right\rangle\left\langle \chi_{\widetilde{\mathrm{T}}_1\nu}\middle|Q_\beta\middle|\chi_{\widetilde{\mathrm{S}}_0\nu}\right\rangle \delta(E_{\widetilde{\mathrm{S}}_0\nu}-E_{\widetilde{\mathrm{T}}_1\nu}). \quad (4)$$

ν: vibrational quantum number, $\chi_{M\nu}$: vibrational wave function of state $M\nu$.

For the transition from T_1–S_1 mixed state to S_0 pure state,

$$\left|\widetilde{\mathrm{T}}_1\right\rangle = C^{\widetilde{\mathrm{T}}_1}_{\mathrm{S}_1}\left|\mathrm{S}_1\right\rangle + C^{\widetilde{\mathrm{T}}_1}_{\mathrm{T}_1}\left|\mathrm{T}_1\right\rangle, \quad \left|\widetilde{\mathrm{S}}_0\right\rangle = \left|\mathrm{S}_0\right\rangle.$$



That's why we have to consider only VCC below:

$$V_\alpha^{\widetilde{\mathrm{T}}_1-\widetilde{\mathrm{S}}_0} = \left(C^{\widetilde{\mathrm{T}}_1}_{\mathrm{S}_1}\right)^* V_\alpha^{\mathrm{S}_1-\mathrm{S}_0}. \quad (5)$$

VCC is written by vibronic coupling density (VCD) $\eta_\alpha^{\mathrm{S}_1-\mathrm{S}_0}(\boldsymbol{r})$:

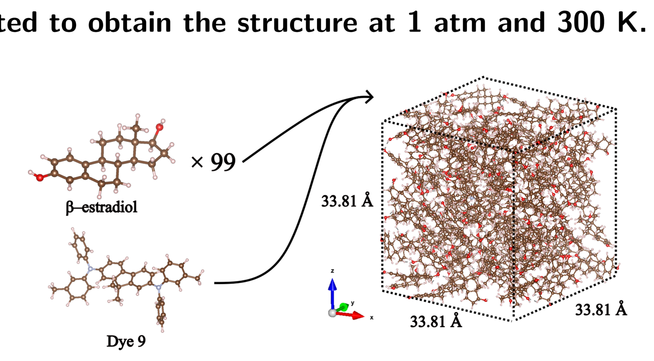
$$V_\alpha^{\mathrm{S}_1-\mathrm{S}_0} = \int \eta_\alpha^{\mathrm{S}_1-\mathrm{S}_0}(\boldsymbol{r})d\boldsymbol{r} = \int \rho^{\mathrm{S}_1-\mathrm{S}_0}(\boldsymbol{r})\nu_\alpha(\boldsymbol{r})d\boldsymbol{r}. \quad (6)$$

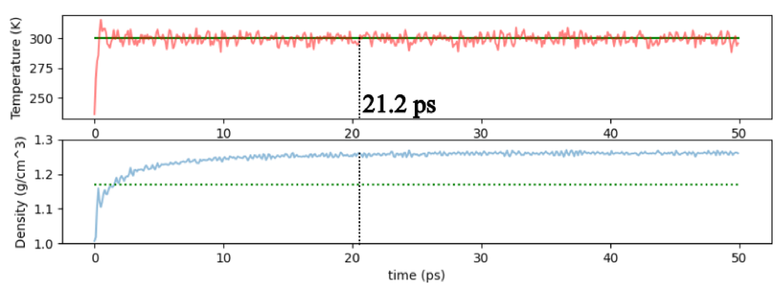
$\rho^{\mathrm{S}_1-\mathrm{S}_0}(\boldsymbol{r})$ is the overlap density and $\nu_\alpha(\boldsymbol{r})$ is the potential derivative.

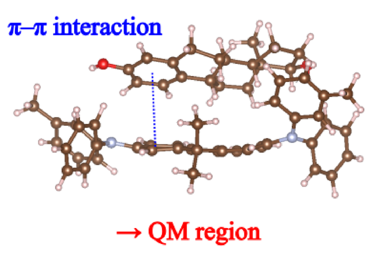
Results: Obtaining the Structure in the β–Estradiol Matrix

NPT–MD simulations are conducted to obtain the structure at 1 atm and 300 K.

- Method: Nosé–Hoover thermostat
- Temperature: 300 K

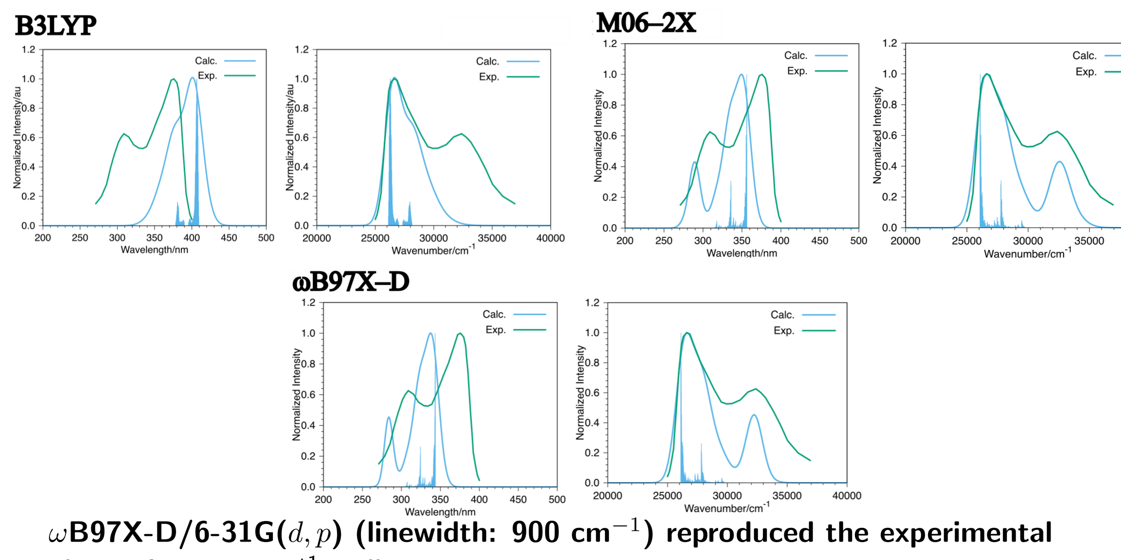






Results: Selection of Functional

Absorption spectra in THF solution[*6] were calculated.

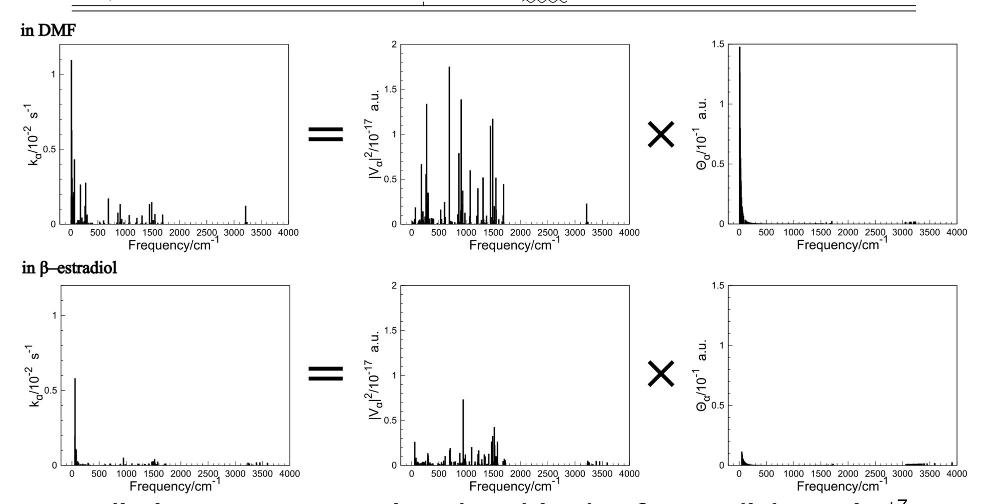


ωB97X-D/6-31G(d,p) (linewidth: 900 cm^{-1}) reproduced the experimental absorption spectra[*1] well.

[*6] The solvent effect was considered through PCM.

Results: Comparison of Nonradiative Rate Constants

	$k^{\widetilde{\mathrm{T}}_1\to\widetilde{\mathrm{S}}_0}_{\mathrm{nr}}$ /10^{-2} s^{-1}	$\sum_\alpha k^{\widetilde{\mathrm{T}}_1\to\widetilde{\mathrm{S}}_0}_{\mathrm{nr},\alpha\alpha}$ /10^{-2} s^{-1}	$\sum_{\alpha,\beta} k^{\widetilde{\mathrm{T}}_1\to\widetilde{\mathrm{S}}_0}_{\mathrm{nr},\alpha\beta}$ /10^{-2} s^{-1}
in DMF	6.48	5.90	0.583
in β–estradiol	3.61	1.90	1.71



The nonradiative rate constant is reduced in the β–estradiol matrix [*7].

[*7] The linewidths are 900 cm^{-1} in DMF solution and 1100 cm^{-1} in the β–estradiol matrix.

Vibrational modes with large rate constants in DMF solution together with their corresponding modes in the β–estradiol matrix are summarized.

Table1: Correspondence of modes and nonradiative rate constants in DMF solution and in the β–estradiol matrix. Only components more than 10% are tabulated.

in DMF			in β–estradiol				
mode α	freq. /cm^{-1}	$k^{\widetilde{\mathrm{T}}_1\to\widetilde{\mathrm{S}}_0}_{\mathrm{nr},\alpha}$ /s^{-1}	mode α	freq. /cm^{-1}	Coefficient	$k^{\widetilde{\mathrm{T}}_1\to\widetilde{\mathrm{S}}_0}_{\mathrm{nr},\alpha}$ /s^{-1}	ratio of $k^{\widetilde{\mathrm{T}}_1\to\widetilde{\mathrm{S}}_0}_{\mathrm{nr},\alpha}$
2	13.7	1.1×10^{-2}	15	123.0	0.46 (21%)	1.8×10^{-4}	6.1×10^{1}
3	19.7	6.3×10^{-3}	22	158.9	−0.50 (25%)	9.1×10^{-5}	1.4×10^{2}
			17	139.2	−0.36 (13%)	5.0×10^{-7}	1.3×10^{4}
14	68.9	4.3×10^{-3}	27	187.7	0.56 (32%)	7.2×10^{-5}	6.0×10^{1}
4	25.5	3.1×10^{-3}	22	158.9	0.45 (20%)	9.1×10^{-5}	6.9×10^{1}

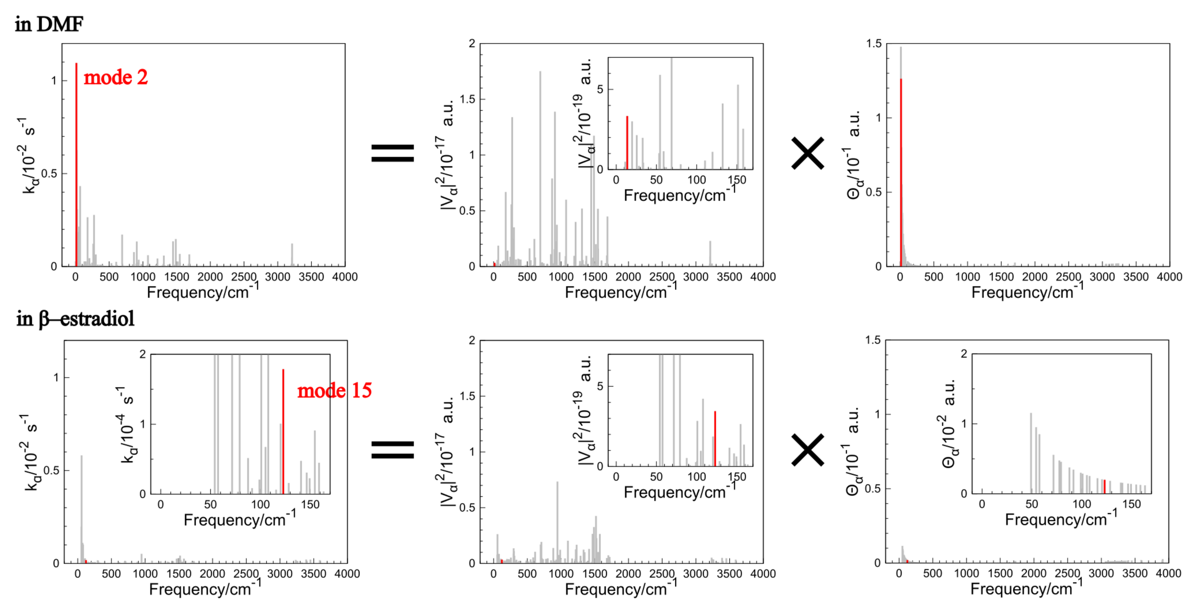
Overall, vibrational frequencies shift to higher wavenumbers in the β–estradiol matr

→ Contributes to the decrease in Θ_α:

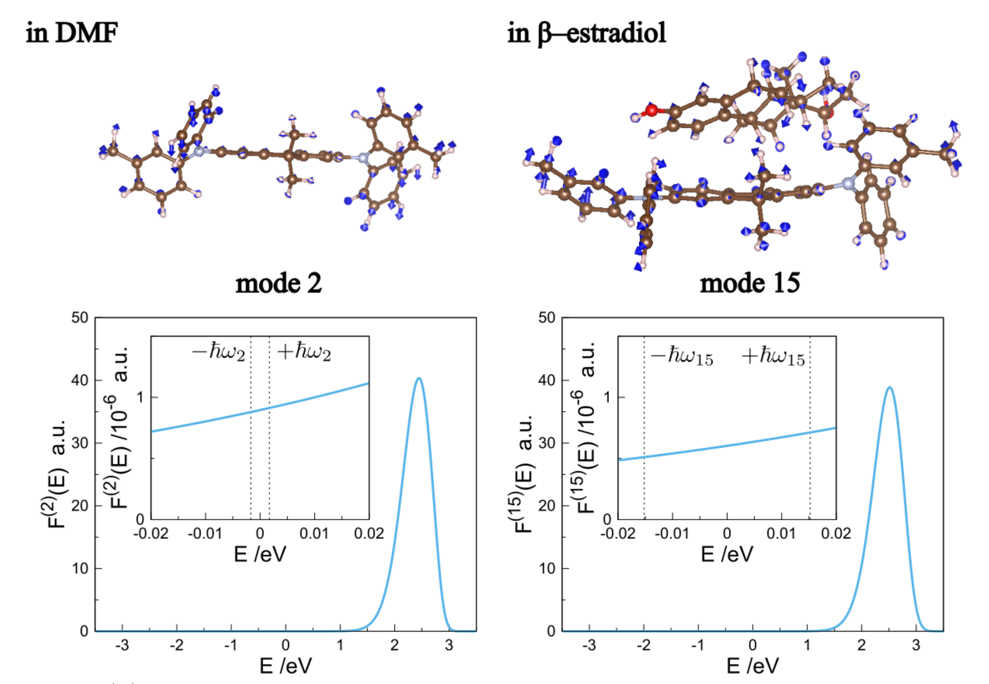
$$\begin{aligned}\Theta_\alpha &:= \Theta_{\alpha\alpha} \\ &= \sum_{\nu_\alpha} P_{\nu_\alpha}(T)\left(\frac{(\nu_\alpha+1)\hbar}{2\omega_\alpha}F^{(\alpha)}(+\hbar\omega_\alpha) + \frac{\nu_\alpha\hbar}{2\omega_\alpha}F^{(\alpha)}(-\hbar\omega_\alpha)\right). \end{aligned} \quad (7)$$

$F^{(\alpha)}(E)$ is called Franck–Condon envelope excluding vibrational mode α.

Results: Mode2 in DMF and Mode15 in β–Estradiol

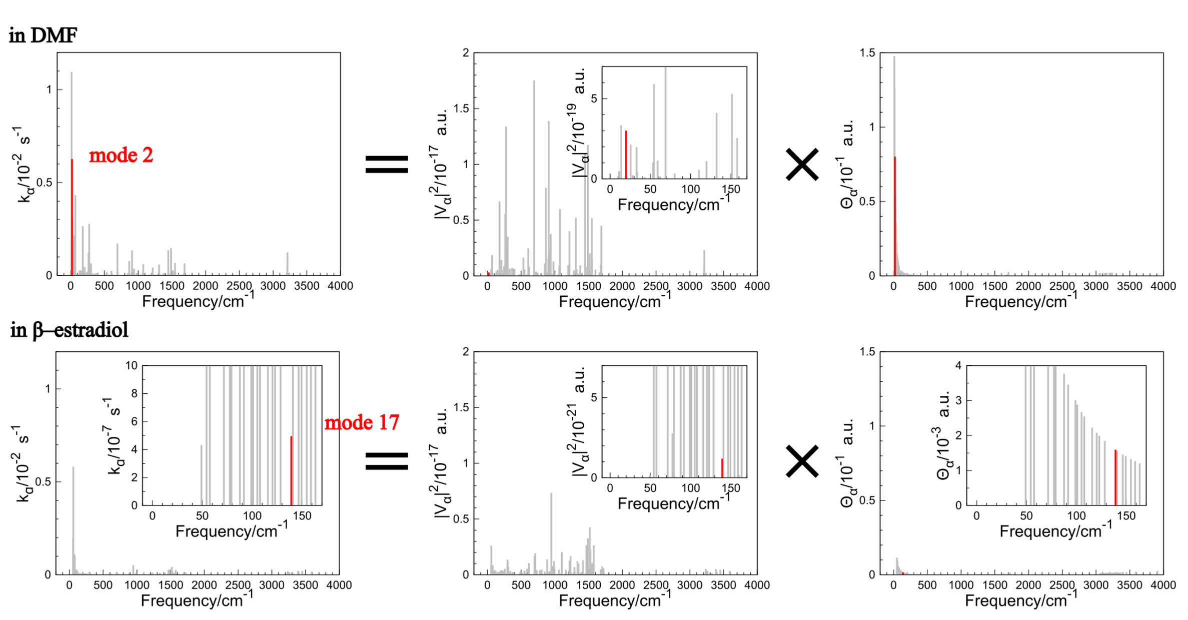


Off–diagonal VCCs are comparable,
whereas the nuclear part Θ_α decreases siginificantly.

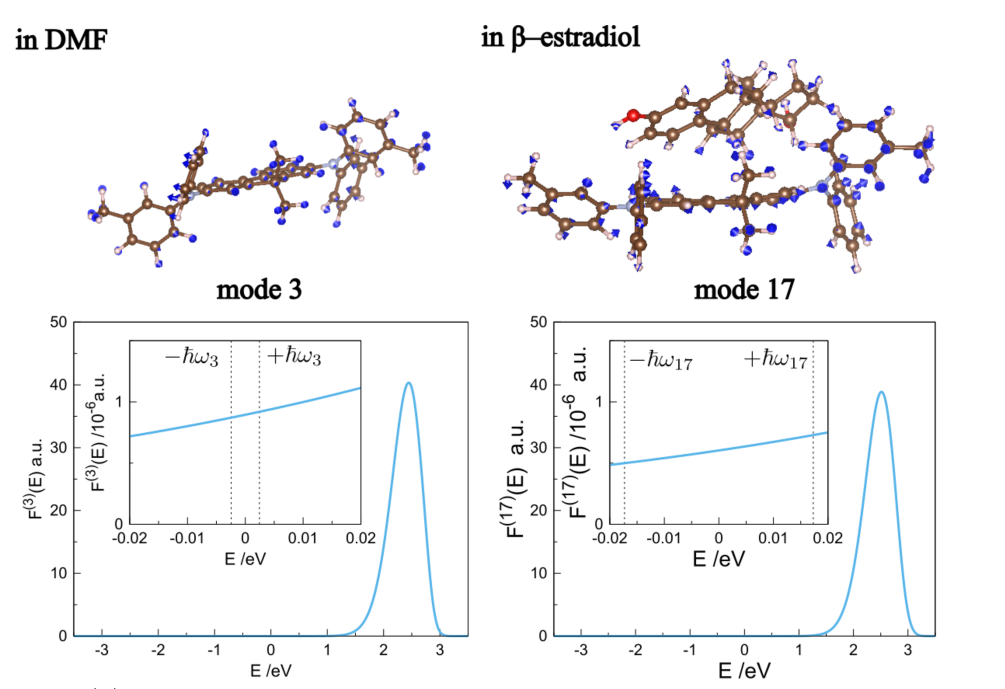


Because $F^{(\alpha)}(E)$ does not decrease significantly,
the decrease of Θ_α is mainly attributed to the increase in the vibrational frequency.

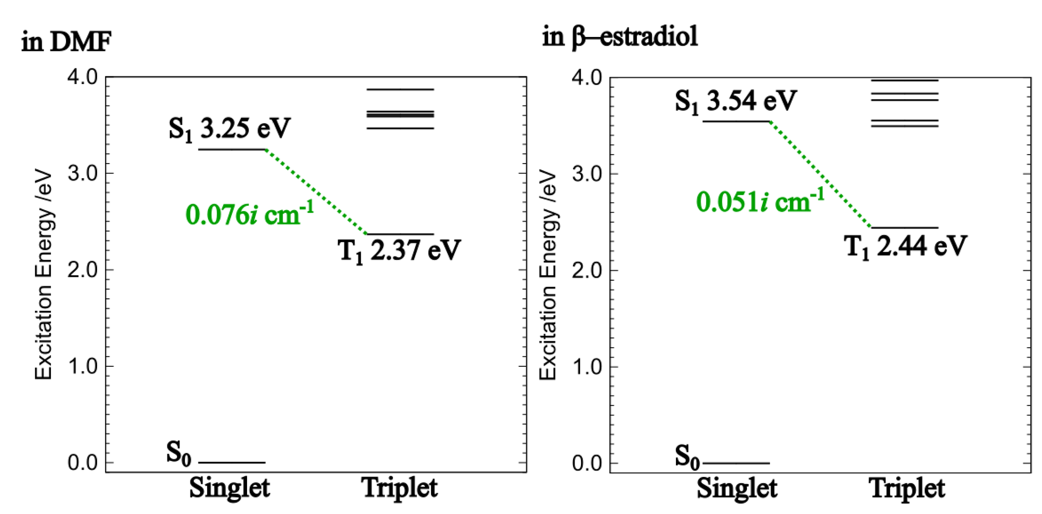
Results: Mode3 in DMF and Mode17 in β–Estradiol



Both off–diagonal VCCs and Θ_α decrease.



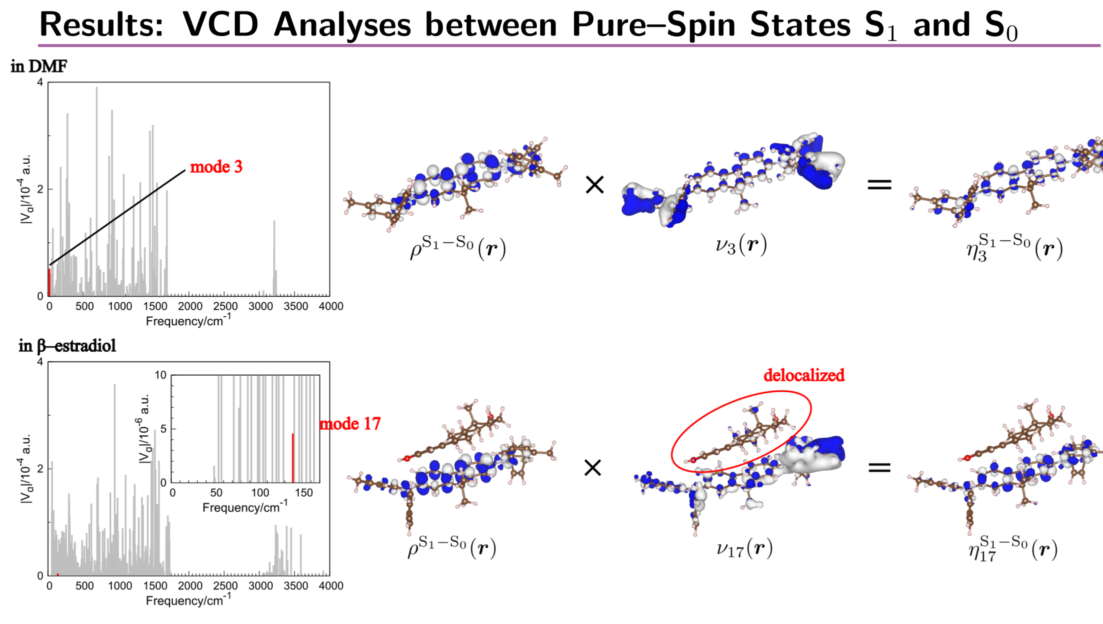
Because $F^{(\alpha)}(E)$ does not decrease significantly,
the decrease of Θ_α is mainly attributed to the increase in the vibrational frequency.



- in DMF solution: $C^{\widetilde{\mathrm{T}}_1}_{\mathrm{S}_1} = -1.1i \times 10^{-5}$
- in the β–estradiol matrix: $C^{\widetilde{\mathrm{T}}_1}_{\mathrm{S}_1} = -7.6i \times 10^{-6}$

The decrease in the magnitude of the mixing coefficient $C^{\widetilde{\mathrm{T}}_1}_{\mathrm{S}_1}$ is attributed to the decrease in off–diagonal VCCs.

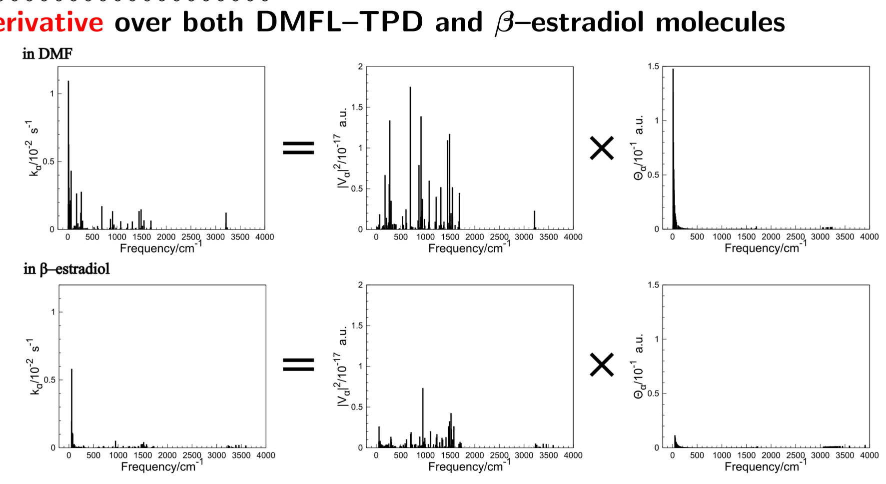
Results: VCD Analyses between Pure–Spin States S_1 and S_0



$\rho^{\mathrm{S}_1-\mathrm{S}_0}(\boldsymbol{r})$ localizes in DMFL–TPD for both systems,
whereas $\nu_{17}(\boldsymbol{r})$ delocalizes both DMFL–TPD and β–estradiol molecules.
→Decrease in off–diagonal VCCs.

Conclusion

- Nonradiative rate constant of DMFL–TPD decreases in the β–estradiol matrix compared to that in DMF solution.
- This reduction is attributed to
  - the decrease in Θ_α caused by the increase of vibrational frequencies
  - the decrease in VCCs caused by the delocalization of the potential derivative over both DMFL–TPD and β–estradiol molecules



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